12.0 ANALOG-TO-DIGITAL CONVERTER (A/D) MODULE

The Analog-to-Digital (A/D) converter module has seven inputs for 18/20 pin devices (PIC16F88 devices only).

The conversion of an analog input signal results in a corresponding 10-bit digital number. The A/D module has a high and low-voltage reference input that is software selectable to some combination of VDD, VSS, VREF- (RA2) or VREF+ (RA3).

The A/D converter has a unique feature of being able to operate while the device is in Sleep mode. To operate in Sleep, the A/D conversion clock must be derived from the A/D's internal RC oscillator.

The A/D module has five registers:

- A/D Result High Register (ADRESH)
- A/D Result Low Register (ADRESL)
- A/D Control Register 0 (ADCON0)
- A/D Control Register 1 (ADCON1)
- Analog Select Register (ANSEL)

The ADCON0 register, shown in Register 12-2, controls the operation of the A/D module. The ANSEL register, shown in Register 12-1 and the ADCON1 register, shown in Register 12-3, configure the functions of the port pins. The port pins can be configured as analog inputs (RA3/RA2 can also be voltage references) or as digital I/O.

Additional information on using the A/D module can be found in the *"PICmicro® Mid-Range MCU Family Reference Manual"* (DS33023).

REGISTER 12-1: ANSEL: ANALOG SELECT REGISTER (ADDRESS 9Bh) PIC16F88 DEVICES ONLY

U-0	R/W-1	R/W-1	R/W-1	R/W-1	R/W-1	R/W-1	R/W-1
—	ANS6	ANS5	ANS4	ANS3	ANS2	ANS1	ANS0
bit 7							bit 0

bit 7 **Unimplemented:** Read as '0'

bit 6-0 **ANS<6:0>:** Analog Input Select bits

Bits select input function on corresponding AN<6:0> pins.

1 = Analog I/O(1,2)

0 = Digital I/O

Note 1: Setting a pin to an analog input disables the digital input buffer. The corresponding TRIS bit should be set to input mode when using pins as analog inputs. Only AN2 is an analog I/O, all other ANx pins are analog inputs.

2: See the block diagrams for the analog I/O pins to see how ANSEL interacts with the CHS bits of the ADCON0 register.

Legend:			
R = Readable bit	W = Writable bit	U = Unimplemented bit, read as '0'	
-n = Value at POR	'1' = Bit is set	'0' = Bit is cleared	x = Bit is unknown

REGISTER 12-2: ADCON0: A/D CONTROL REGISTER (ADDRESS 1Fh) PIC16F88 DEVICES ONLY

R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	R/W-0	U-0	R/W-0
ADCS1	ADCS0	CHS2	CHS1	CHS0	GO/$\overline{\text{DONE}}$	—	ADON
bit 7							bit 0

bit 7-6 **ADCS<1:0>:** A/D Conversion Clock Select bits

If ADCS2 = 0:
00 = FOSC/2
01 = FOSC/8
10 = FOSC/32
11 = FRC (clock derived from the internal A/D module RC oscillator)

If ADCS2 = 1:
00 = FOSC/4
01 = FOSC/16
10 = FOSC/64
11 = FRC (clock derived from the internal A/D module RC oscillator)

bit 5-3 **CHS<2:0>:** Analog Channel Select bits

000 = Channel 0 (RA0/AN0)
001 = Channel 1 (RA1/AN1)
010 = Channel 2 (RA2/AN2)
011 = Channel 3 (RA3/AN3)
100 = Channel 4 (RA4/AN4)
101 = Channel 5 (RB6/AN5)
110 = Channel 6 (RB7/AN6)

bit 2 **GO/$\overline{\text{DONE}}$:** A/D Conversion Status bit

If ADON = 1:
1 = A/D conversion in progress (setting this bit starts the A/D conversion)
0 = A/D conversion not in progress (this bit is automatically cleared by hardware when the A/D conversion is complete)

bit 1 **Unimplemented:** Read as '0'

bit 0 **ADON:** A/D On bit

1 = A/D converter module is operating
0 = A/D converter module is shut off and consumes no operating current

Legend:			
R = Readable bit	W = Writable bit	U = Unimplemented bit, read as '0'	
-n = Value at POR	'1' = Bit is set	'0' = Bit is cleared	x = Bit is unknown

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REGISTER 12-3: ADCON1: A/D CONTROL REGISTER 1 (ADDRESS 9Fh) PIC16F88 DEVICES ONLY

R/W-0	R/W-0	R/W-0	R/W-0	U-0	U-0	U-0	U-0
ADFM	ADCS2	VCFG1	VCFG0	—	—	—	—
bit 7							bit 0

bit 7 **ADFM:** A/D Result Format Select bit

1 = Right justified. Six Most Significant bits of ADRESH are read as '0'.
0 = Left justified. Six Least Significant bits of ADRESL are read as '0'.

bit 6 **ADCS2:** A/D Clock Divide by 2 Select bit

1 = A/D clock source is divided by 2 when system clock is used
0 = Disabled

bit 5-4 **VCFG<1:0>:** A/D Voltage Reference Configuration bits

Logic State	VREF+	VREF-
00	AVDD	AVSS
01	AVDD	VREF-
10	VREF+	AVSS
11	VREF+	VREF-

Note: The ANSEL bits for AN3 and AN2 inputs must be configured as analog inputs for the VREF+ and VREF- external pins to be used.

bit 3-0 **Unimplemented:** Read as '0'

Legend:			
R = Readable bit	W = Writable bit	U = Unimplemented bit, read as '0'	
-n = Value at POR	'1' = Bit is set	'0' = Bit is cleared	x = Bit is unknown

The ADRESH:ADRESL registers contain the result of the A/D conversion. When the A/D conversion is complete, the result is loaded into the A/D Result register pair, the GO/$\overline{\text{DONE}}$ bit (ADCON0<2>) is cleared and A/D Interrupt Flag bit, ADIF, is set. The block diagram of the A/D module is shown in Figure 12-1.

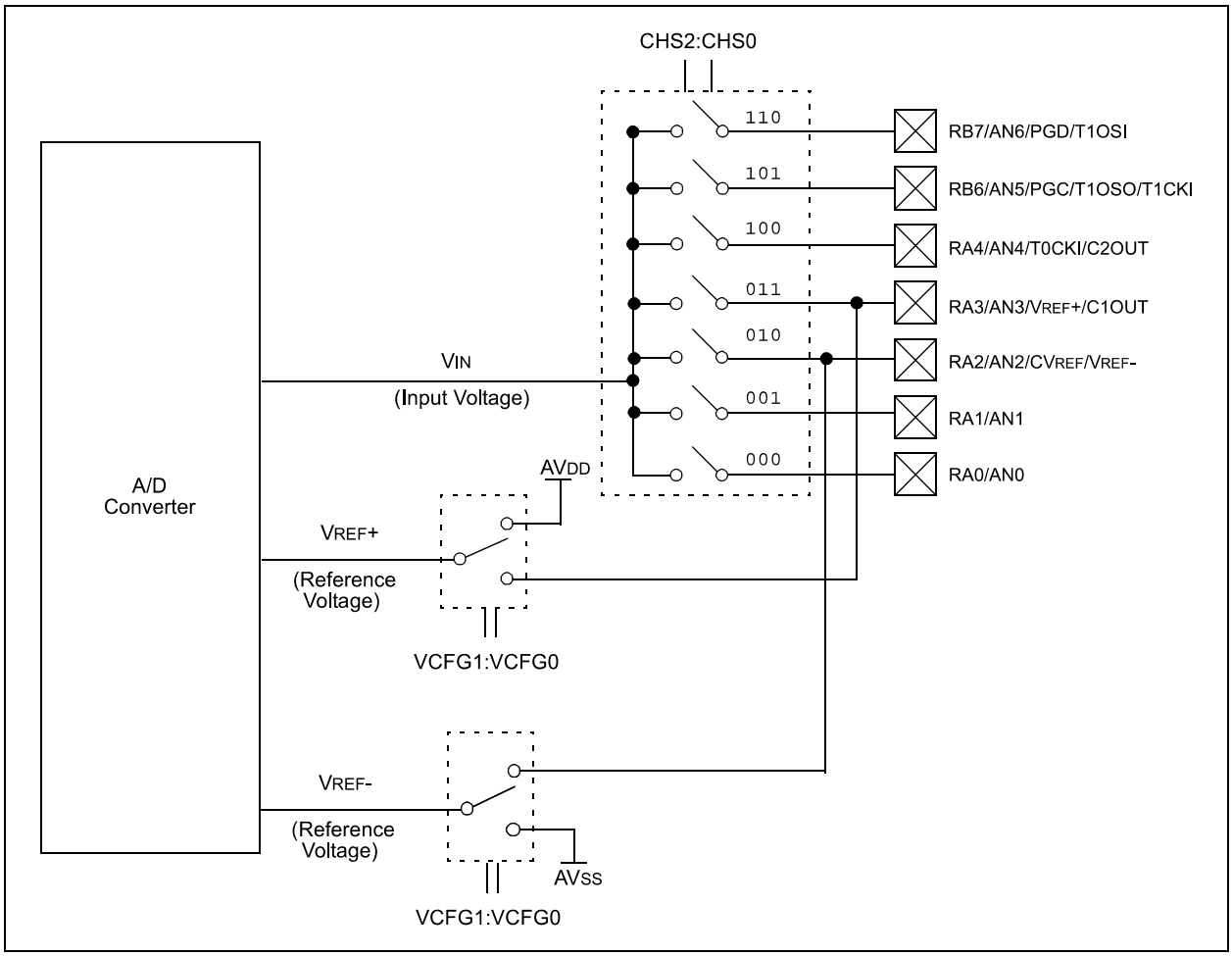
After the A/D module has been configured as desired, the selected channel must be acquired before the conversion is started. The analog input channels must have their corresponding TRIS bits selected as inputs.

To determine sample time, see **Section 12.1 "A/D Acquisition Requirements"**. After this sample time has elapsed, the A/D conversion can be started.

These steps should be followed for doing an A/D conversion:

1. Configure the A/D module:
   - Configure analog/digital I/O (ANSEL)
   - Configure voltage reference (ADCON1)
   - Select A/D input channel (ADCON0)
   - Select A/D conversion clock (ADCON0)
   - Turn on A/D module (ADCON0)
2. Configure A/D interrupt (if desired):
   - Clear ADIF bit
   - Set ADIE bit
   - SET PEIE bit
   - Set GIE bit
3. Wait the required acquisition time.
4. Start conversion:
   - Set GO/$\overline{\text{DONE}}$ bit (ADCON0)
5. Wait for A/D conversion to complete, by either:
   - Polling for the GO/$\overline{\text{DONE}}$ bit to be cleared (with interrupts disabled); OR
   - Waiting for the A/D interrupt
6. Read A/D Result register pair (ADRESH:ADRESL), clear bit ADIF if required.
7. For next conversion, go to step 1 or step 2 as required. The A/D conversion time per bit is defined as T_{AD}. A minimum wait of 2 T_{AD} is required before the next acquisition starts.

FIGURE 12-1: A/D BLOCK DIAGRAM



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12.1 A/D Acquisition Requirements

For the A/D converter to meet its specified accuracy, the charge holding capacitor (CHOLD) must be allowed to fully charge to the input channel voltage level. The analog input model is shown in Figure 12-2. The source impedance (RS) and the internal sampling switch (RSS) impedance directly affect the time required to charge the capacitor CHOLD. The sampling switch (RSS) impedance varies over the device voltage (VDD), see Figure 12-2. **The maximum recommended impedance for analog sources is 10 kΩ.** As the impedance is decreased, the acquisition time may be decreased. After the analog input channel is selected (changed), this acquisition must be done before the conversion can be started.

To calculate the minimum acquisition time, Equation 12-1 may be used. This equation assumes that 1/2 LSb error is used (1024 steps for the A/D). The 1/2 LSb error is the maximum error allowed for the A/D to meet its specified resolution.

To calculate the minimum acquisition time, TACQ, see the *"PICmicro® Mid-Range MCU Family Reference Manual"* (DS33023).

EQUATION 12-1: ACQUISITION TIME

TACQ = Amplifier Settling Time + Hold Capacitor Charging Time + Temperature Coefficient

= TAMP + TC + TCOFF
= 2 µs + TC + [(Temperature -25°C)(0.05 µs/°C)]
TC = CHOLD (RIC + RSS + RS) In(1/2047)
= -120 pF (1 kΩ + 7 kΩ + 10 kΩ) In(0.0004885)
= 16.47 µs
TACQ = 2 µs + 16.47 µs + [(50°C – 25°C)(0.05 µs/°C)
= 19.72 µs

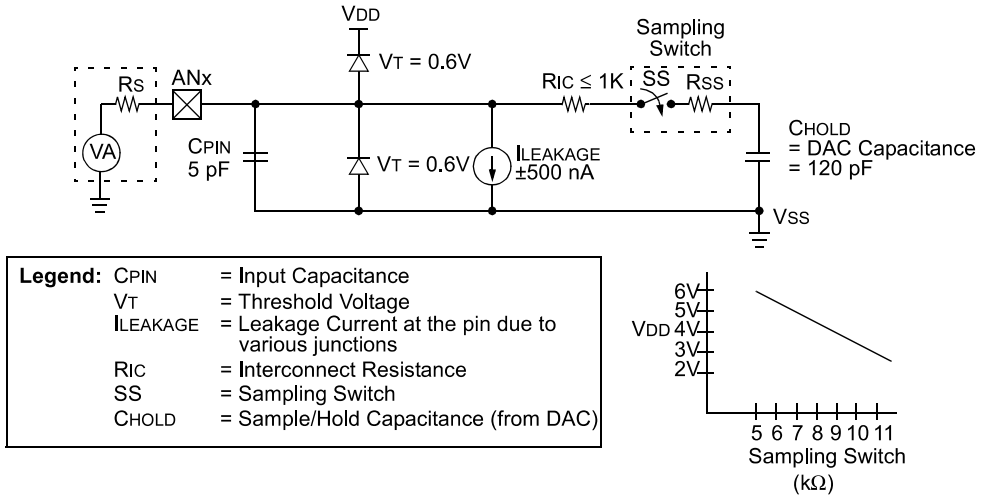
Note 1: The reference voltage (VREF) has no effect on the equation, since it cancels itself out.

2: The charge holding capacitor (CHOLD) is not discharged after each conversion.

3: The maximum recommended impedance for analog sources is 10 kΩ. This is required to meet the pin leakage specification.

4: After a conversion has completed, a 2.0 TAD delay must complete before acquisition can begin again. During this time, the holding capacitor is not connected to the selected A/D input channel.

FIGURE 12-2: ANALOG INPUT MODEL



12.2 Selecting the A/D Conversion Clock

The A/D conversion time per bit is defined as TAD. The A/D conversion requires 9.0 TAD per 10-bit conversion. The source of the A/D conversion clock is software selectable. The seven possible options for TAD are:

- 2 TOSC
- 4 TOSC
- 8 TOSC
- 16 TOSC
- 32 TOSC
- 64 TOSC
- Internal A/D module RC oscillator (2-6 μs)

For correct A/D conversions, the A/D conversion clock (TAD) must be selected to ensure a minimum TAD time as small as possible, but no less than 1.6 μs and not greater than 6.4 μs.

Table 12-1 shows the resultant TAD times derived from the device operating frequencies and the A/D clock source selected.

12.3 Operation in Power-Managed Modes

The selection of the automatic acquisition time and A/D conversion clock is determined in part by the clock source and frequency while in a power-managed mode.

If the A/D is expected to operate while the device is in a power-managed mode, the ADCS2:ADCS0 bits in ADCON0 and ADCON1 should be updated in accordance with the power-managed mode clock that will be used. After the power-managed mode is entered (either of the power-managed Run modes), an A/D acquisition or conversion may be started. Once an acquisition or conversion is started, the device should continue to be clocked by the same power-managed mode clock source until the conversion has been completed.

If the power-managed mode clock frequency is less than 1 MHz, the A/D RC clock source should be selected.

TABLE 12-1: TAD vs. MAXIMUM DEVICE OPERATING FREQUENCIES – STANDARD DEVICES (C)

AD Clock Source (TAD)			Maximum Device Frequency
Operation	**ADCS<2>**	**ADCS<1:0>**	**Max.**
2 TOSC	0	00	1.25 MHz
4 TOSC	1	00	2.5 MHz
8 TOSC	0	01	5 MHz
16 TOSC	1	01	10 MHz
32 TOSC	0	10	20 MHz
64 TOSC	1	10	20 MHz
RC(1,2,3)	x	11	**(Note 1)**

Note 1: The RC source has a typical TAD time of 4 μs, but can vary between 2-6 μs.

2: When the device frequencies are greater than 1 MHz, the RC A/D conversion clock source is only recommended for Sleep operation.

3: For extended voltage devices (LF), please refer to **Section 18.0 "Electrical Characteristics"**.

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12.4 Configuring Analog Port Pins

The ADCON1, ANSEL, TRISA and TRISB registers control the operation of the A/D port pins. The port pins that are desired as analog inputs must have their corresponding TRIS bits set (input). If the TRIS bit is cleared (output), the digital output level (V_{OH} or V_{OL}) will be converted.

The A/D operation is independent of the state of the CHS<2:0> bits and the TRIS bits.

Note 1:	When reading the Port register, all pins configured as analog input channels will read as cleared (a low level). Pins configured as digital inputs will convert an analog input. Analog levels on a digitally configured input will not affect the conversion accuracy.
2:	Analog levels on any pin that is defined as a digital input (including the RA4:RA0 and RB7:RB6 pins), may cause the input buffer to consume current out of the device specification.

12.5 A/D Conversions

Clearing the GO/$\overline{\text{DONE}}$ bit during a conversion will abort the current conversion. The A/D Result register pair will NOT be updated with the partially completed A/D conversion sample. That is, the ADRESH:ADRESL registers will continue to contain the value of the last completed conversion (or the last value written to the ADRESH:ADRESL registers). After the A/D conversion is aborted, a 2 T_{AD} wait is required before the next acquisition is started. After this 2 T_{AD} wait, acquisition on the selected channel is automatically started. The GO/$\overline{\text{DONE}}$ bit can then be set to start the conversion.

In Figure 12-3, after the GO/$\overline{\text{DONE}}$ bit is set, the first time segment has a minimum of T_{CY} and a maximum of T_{AD}.

Note:	The GO/$\overline{\text{DONE}}$ bit should **NOT** be set in the same instruction that turns on the A/D.

12.5.1 A/D RESULT REGISTERS

The ADRESH:ADRESL register pair is the location where the 10-bit A/D result is loaded at the completion of the A/D conversion. This register pair is 16 bits wide. The A/D module gives the flexibility to left or right justify the 10-bit result in the 16-bit result register. The A/D Format Select bit (ADFM) controls this justification. Figure 12-4 shows the operation of the A/D result justification. The extra bits are loaded with '0's. When an A/D result will not overwrite these locations (A/D disable), these registers may be used as two general purpose 8-bit registers.

FIGURE 12-3: A/D CONVERSION T_{AD} CYCLES

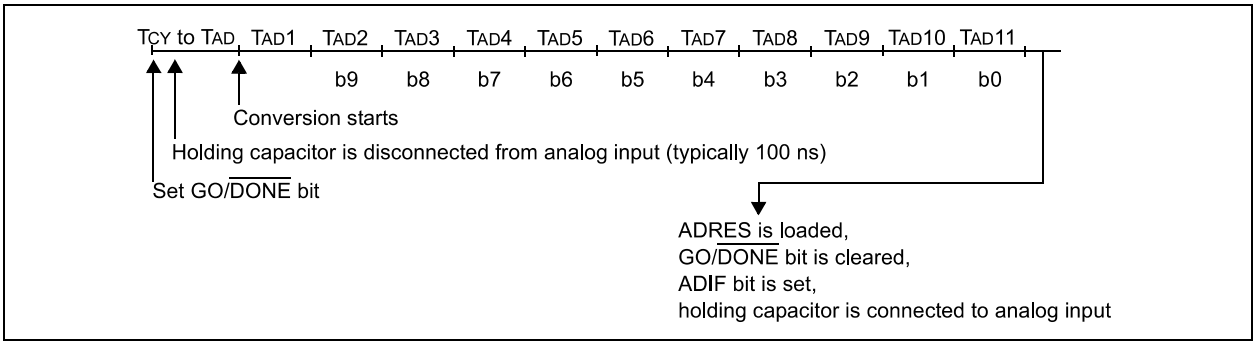
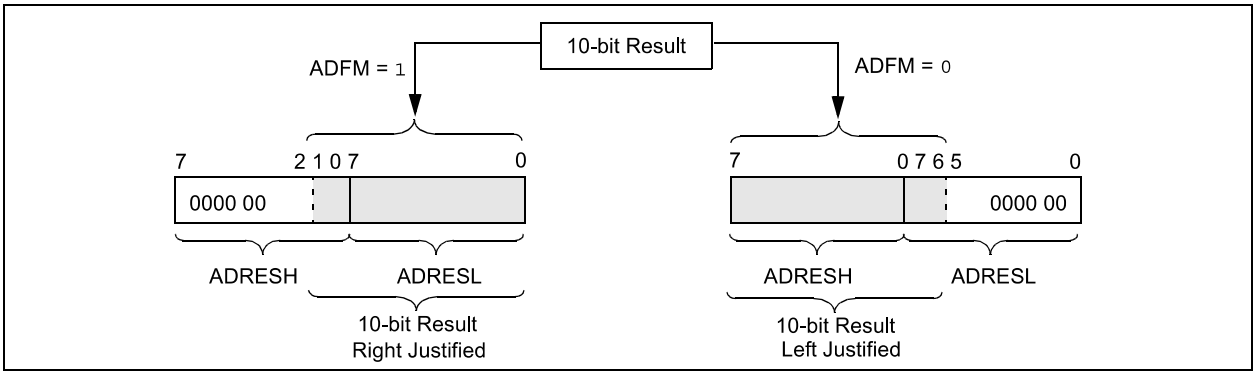


FIGURE 12-4: A/D RESULT JUSTIFICATION



12.6 A/D Operation During Sleep

The A/D module can operate during Sleep mode. This requires that the A/D clock source be set to RC (ADCS1:ADCS0 = 11). When the RC clock source is selected, the A/D module waits one instruction cycle before starting the conversion. This allows the SLEEP instruction to be executed which eliminates all digital switching noise from the conversion. When the conversion is completed, the GO/$\overline{\text{DONE}}$ bit will be cleared and the result loaded into the ADRES registers. If the A/D interrupt is enabled, the device will wake-up from Sleep. If the A/D interrupt is not enabled, the A/D module will then be turned off, although the ADON bit will remain set.

When the A/D clock source is another clock option (not RC), a SLEEP instruction will cause the present conversion to be aborted and the A/D module to be turned off, though the ADON bit will remain set.

Turning off the A/D places the A/D module in its lowest current consumption state.

Note:	For the A/D module to operate in Sleep, the A/D clock source must be set to RC (ADCS1:ADCS0 = 11). To perform an A/D conversion in Sleep, ensure the SLEEP instruction immediately follows the instruction that sets the GO/$\overline{\text{DONE}}$ bit.

12.7 Effects of a Reset

A device Reset forces all registers to their Reset state. The A/D module is disabled and any conversion in progress is aborted. All A/D input pins are configured as analog inputs.

The value that is in the ADRESH:ADRESL registers is not modified for a Power-on Reset. The ADRESH:ADRESL registers will contain unknown data after a Power-on Reset.

12.8 Use of the CCP Trigger

An A/D conversion can be started by the "special event trigger" of the CCP module. This requires that the CCP1M3:CCP1M0 bits (CCP1CON<3:0>) be programmed as '1011' and that the A/D module is enabled (ADON bit is set). When the trigger occurs, the GO/$\overline{\text{DONE}}$ bit will be set, starting the A/D conversion and the Timer1 counter will be reset to zero. Timer1 is reset to automatically repeat the A/D acquisition period with minimal software overhead (moving the ADRESH:ADRESL to the desired location). The appropriate analog input channel must be selected and the minimum acquisition done before the "special event trigger" sets the GO/$\overline{\text{DONE}}$ bit (starts a conversion).

If the A/D module is not enabled (ADON is cleared), then the "special event trigger" will be ignored by the A/D module, but will still reset the Timer1 counter.

TABLE 12-2: REGISTERS/BITS ASSOCIATED WITH A/D

Address	Name	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	Value on POR, BOR	Value on all other Resets
0Bh, 8Bh 10Bh, 18Bh	INTCON	GIE	PEIE	TMR0IE	INT0IE	RBIE	TMR0IF	INT0IF	RBIF	0000 000x	0000 000u
0Ch	PIR1	—	ADIF(1)	RCIF	TXIF	SSPIF	CCP1IF	TMR2IF	TMR1IF	-000 0000	-000 0000
8Ch	PIE1	—	ADIE(1)	RCIE	TXIE	SSPIE	CCP1IE	TMR2IE	TMR1IE	-000 0000	-000 0000
1Eh	ADRESH(2)	A/D Result Register High Byte								xxxx xxxx	uuuu uuuu
9Eh	ADRESL(2)	A/D Result Register Low Byte								xxxx xxxx	uuuu uuuu
1Fh	ADCON0(2)	ADCS1	ADCS0	CHS2	CHS1	CHS0	GO/$\overline{\text{DONE}}$	—	ADON	0000 00-0	0000 00-0
9Fh	ADCON1(2)	ADFM	ADCS2	VCFG1	VCFG0	—	—	—	—	0000 ----	0000 ----
9Bh	ANSEL(2	—	ANS6	ANS5	ANS4	ANS3	ANS2	ANS1	ANS0	-111 1111	-111 1111
05h	PORTA (PIC16F87) (PIC16F88)	RA7	RA6	RA5	RA4	RA3	RA2	RA1	RA0	xxxx 0000 xxx0 0000	uuuu 0000 uuu0 0000
05h, 106h	PORTB (PIC16F87) (PIC16F88)	RB7	RB6	RB5	RB4	RB3	RB2	RB1	RB0	xxxx xxxx 00xx xxxx	uuuu uuuu 00uu uuuu
85h	TRISA	TRISA7	TRISA6	TRISA5(3)	PORTA Data Direction Register (TRISA<4:0>)					1111 1111	1111 1111
86h, 186h	TRISB	TRISB7	TRISB6	TRISB5	TRISB4	TRISB3	TRISB2	TRISB1	TRISB0	1111 1111	1111 1111

Legend: x = unknown, u = unchanged, - = unimplemented, read as '0'. Shaded cells are not used for A/D conversion.

Note 1: This bit is only implemented on the PIC16F88. The bit will read '0' on the PIC16F87.

2: PIC16F88 only.

3: Pin 5 is an input only; the state of the TRISA5 bit has no effect and will always read '1'.

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